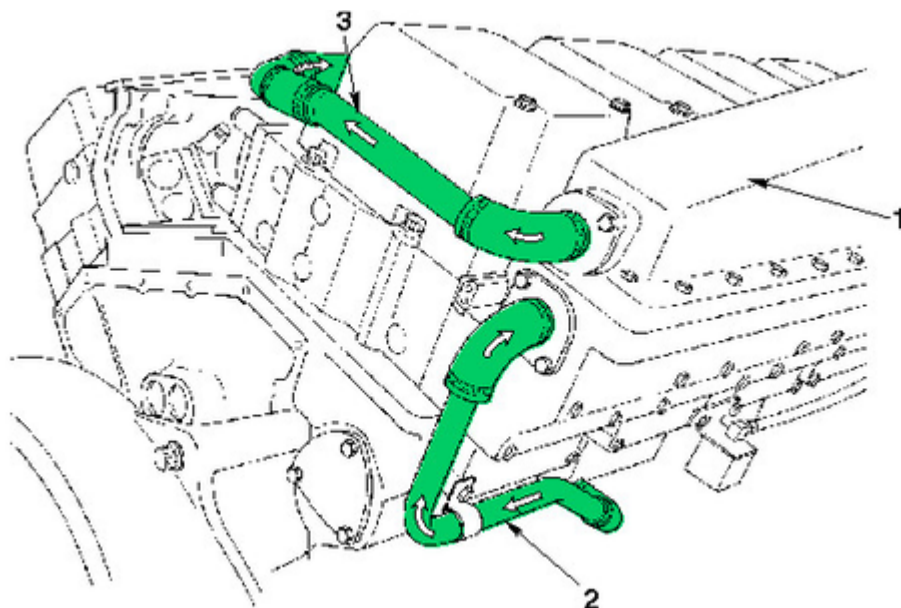


Flow Diagram

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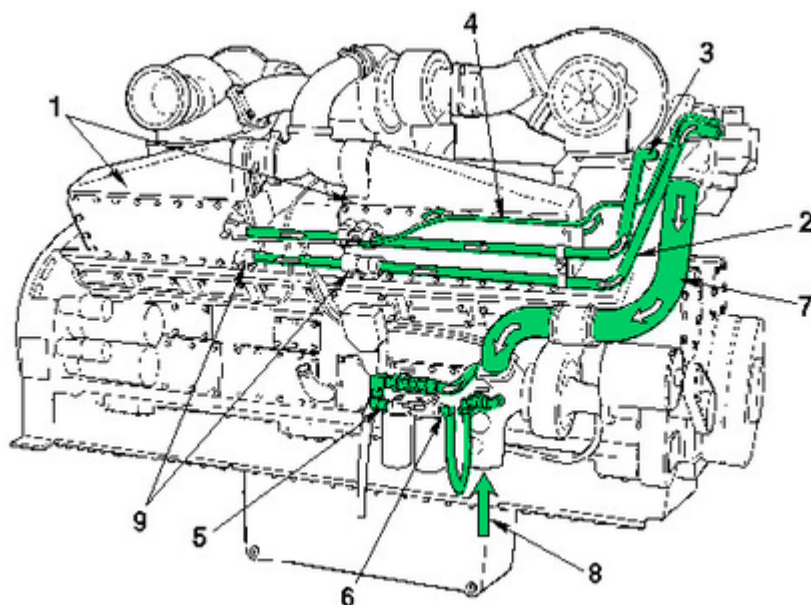


09600613

Outboard (Side Mounted) Aftercooler

1. Aftercooler housing
2. Aftercooler coolant supply
3. Aftercooler coolant return.

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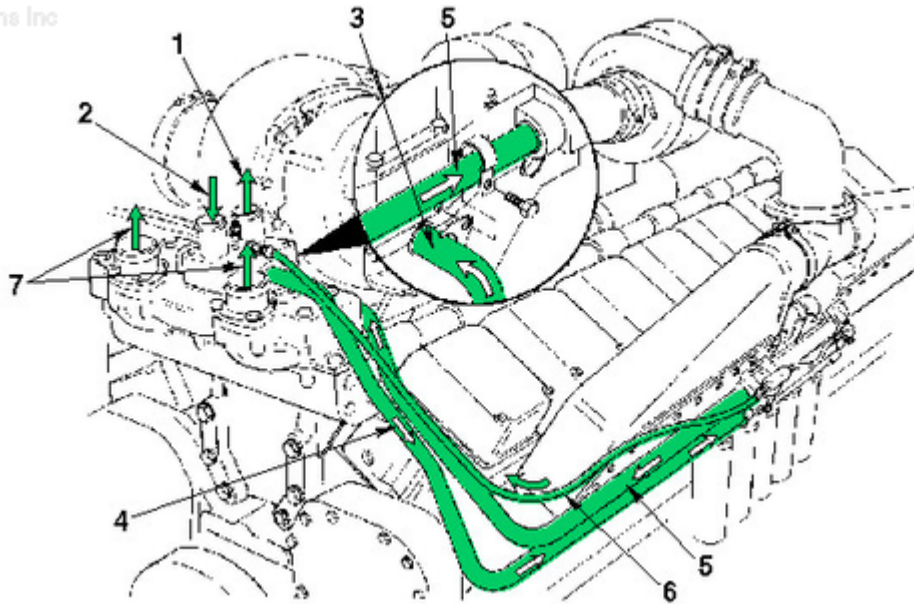


09600614

Low Temperature Aftercooler

1. Aftercooler housing
2. Aftercooler coolant supply
3. Aftercooler coolant return
4. Aftercooler core vent
5. Coolant filter inlet
6. Coolant filter outlet
7. Bypass tube
8. Coolant supply from radiator
9. Aftercooler core drain.

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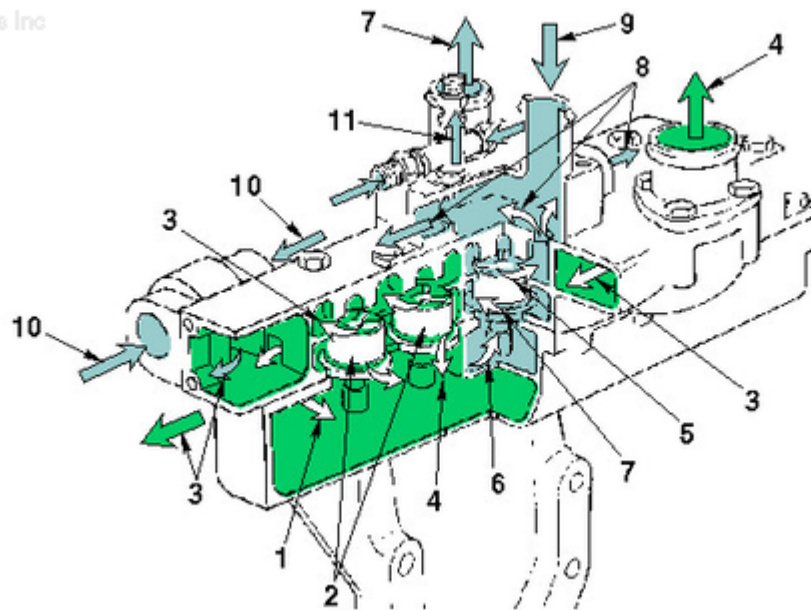


09600616

Thermostat Housing Flow

1. To low temperature aftercooler radiator
2. From low temperature aftercooler radiator
3. To low temperature aftercooler thermostat
4. Aftercooler coolant supply
5. Aftercooler coolant return
6. Aftercooler core vent
7. Coolant return to radiator.

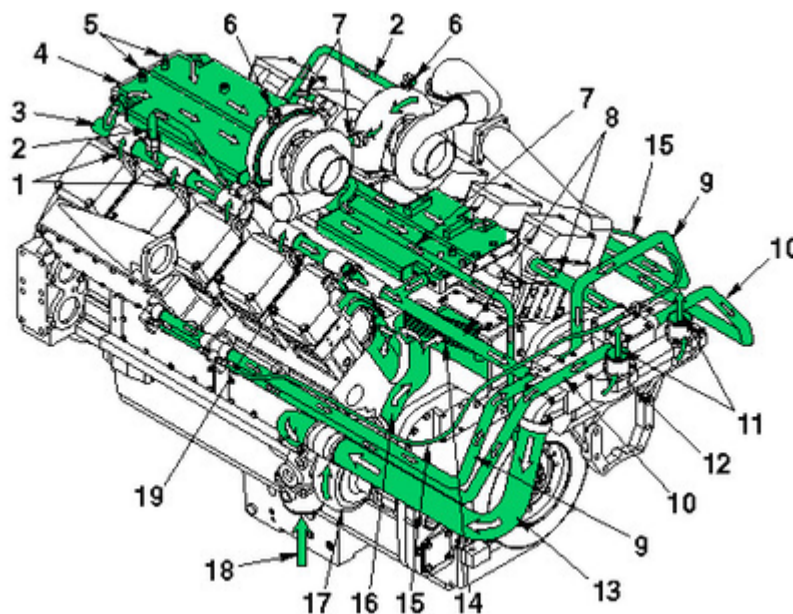
Note : The KTA50-G3/G4 and KTTA50-G2 engines contain low temperature aftercooler hardware, but are **not** low-temperature aftercooled. These engines do **not** flow coolant to or from a low temperature aftercooler radiator, items (1) or (2).



09800614

Low Temperature Aftercooler Thermostat Housing - Marine

1. From coolant manifold
2. Main engine thermostats
3. Bypass flow to coolant pump
4. Coolant to engine heat exchanger or keel cooler
5. Low temperature aftercooler thermostat
6. Coolant supply from vee-cavity
7. To low temperature aftercooler heat exchanger or keel cooler
8. To aftercoolers
9. From low temperature aftercooler heat exchanger or keel cooler
10. From aftercoolers
11. Vent to expansion tank.



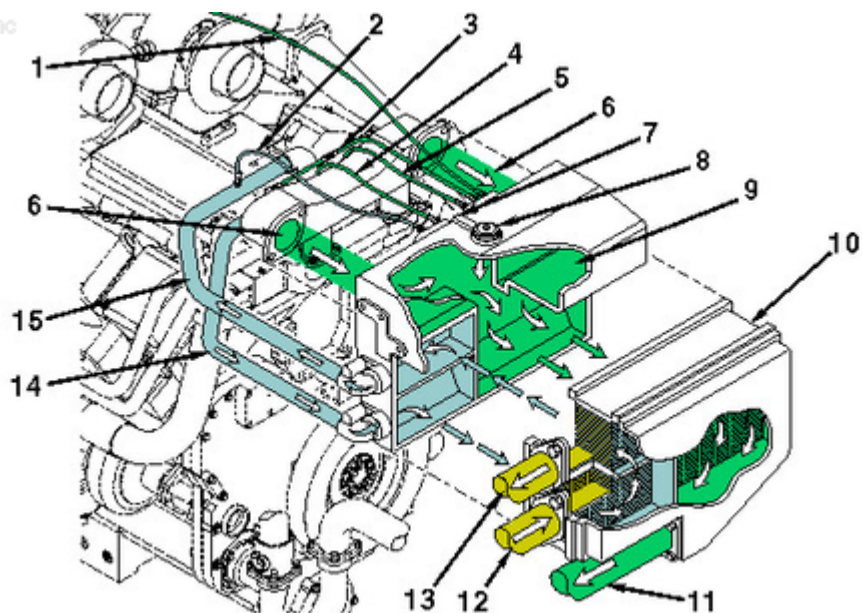
09800607

K38M Keel Cooled (non-low temperature aftercooler)

1. Coolant flow from block

2. Turbocharger coolant supply
3. Rear transfer tube
4. Heat shield
5. Heat shield vent
6. Turbocharger vent to expansion tank
7. Turbocharger coolant return to bypass
8. Coolant return to thermostat housing
9. Aftercooler coolant return
10. Aftercooler coolant supply
11. Coolant outlet to keel cooler
12. Thermostat housing
13. Coolant bypass to coolant pump
14. Lubricating oil cooler
15. Aftercooler vent line
16. Coolant supply to block
17. Coolant pump
18. Coolant inlet from keel cooler
19. Front transfer tube.

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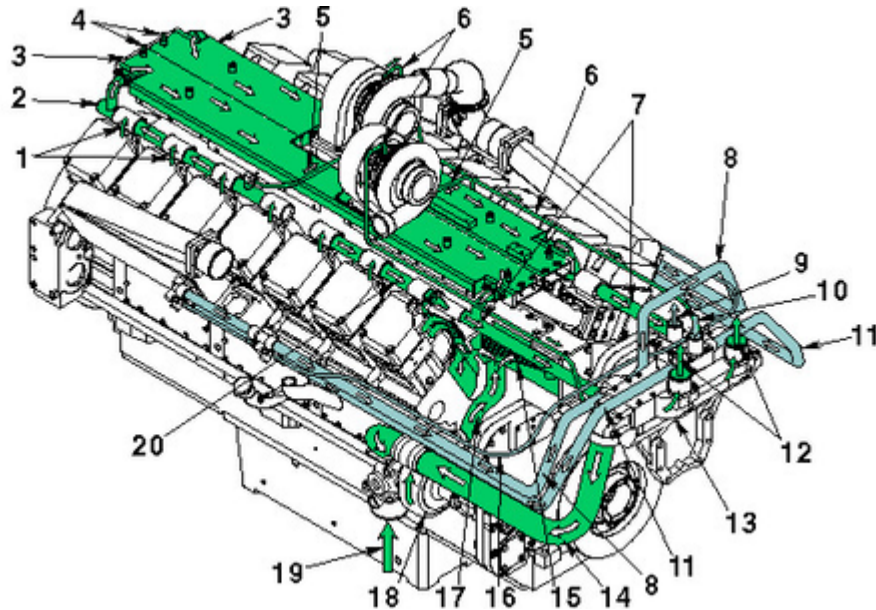
03800605

K38M and K50M Heat Exchanger - Shell and Tube

1. Turbocharger vent line
2. Low temperature aftercooler vent line
3. Coolant connection vent line
4. Thermostat housing and aftercooler vent line
5. Heat shield vent line
6. Engine coolant from thermostat housing
7. Pressure cap overflow vent
8. Pressure cap
9. Expansion tank
10. Heat exchanger

11. Engine coolant outlet to coolant pump
12. Seawater inlet from seawater pump
13. Seawater outlet
14. Low temperature aftercooler coolant from thermostat housing
15. Low temperature aftercooler coolant to thermostat housing.

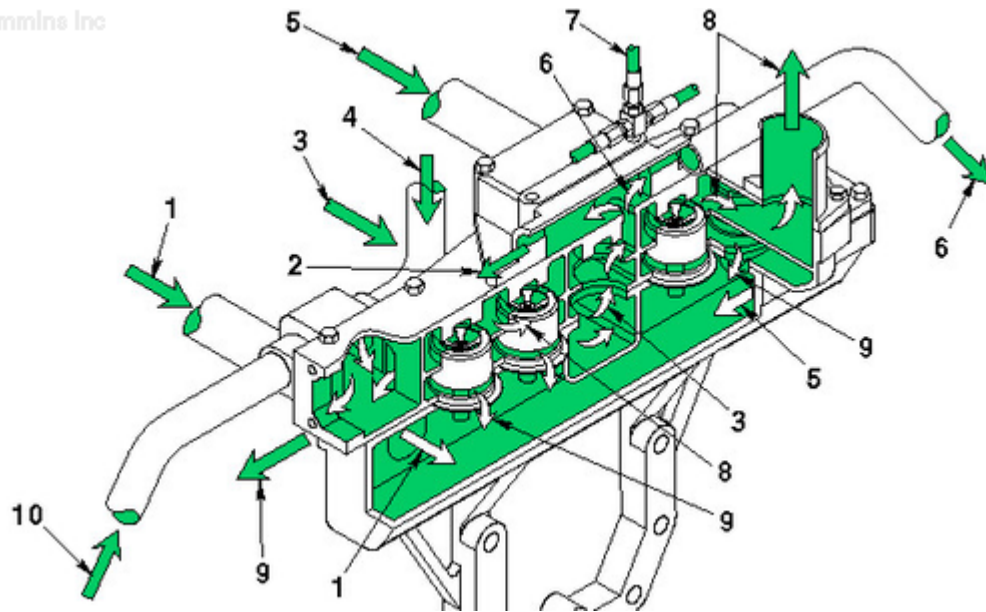
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02800609

K50 Keel Cooled (Low Temperature Aftercooler)

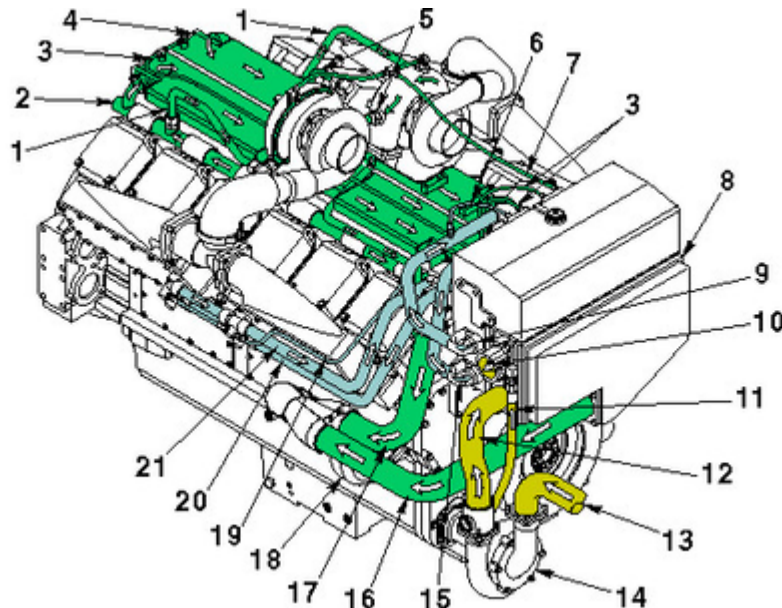
1. Coolant flow from block to transfer tube
2. Rear transfer tube
3. Heat shields
4. Heat shield vents
5. Coolant supply to turbocharger bearing housing
6. Turbocharger bearing housing coolant return
7. Coolant return to thermostat housing
8. Aftercooler coolant return
9. Low temperature aftercooler coolant outlet to keel cooler
10. Low temperature aftercooler coolant inlet from keel cooler
11. Aftercooler coolant supply
12. Coolant outlet to keel cooler
13. Thermostat housing
14. Coolant bypass to coolant pump
15. Lubricating oil cooler
16. Aftercooler vent line
17. Coolant supply to block
18. Coolant pump
19. Coolant inlet from keel cooler
20. Front transfer tube.



09600610

Non-Low Temperature Aftercooler Thermostat Housing - Marine

1. From right bank transfer tube
2. To right bank aftercooler
3. From engine center vee-cavity
4. From left bank aftercooler
5. From left bank transfer tube
6. To left bank aftercooler
7. Vent to expansion tank
8. To keel cooler
9. Coolant bypass
10. From right bank aftercooler.



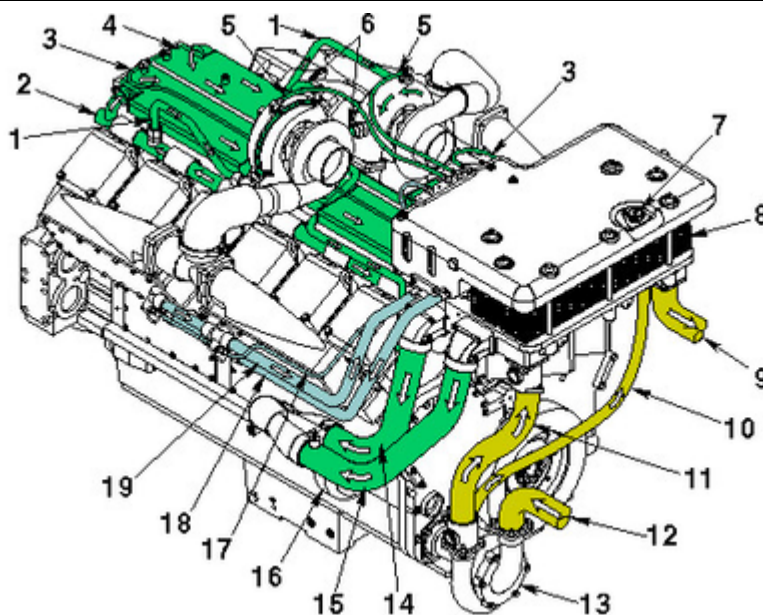
09600611

K38M Heat Exchanger Low Temperature Aftercooler - Shell and Tube

1. Turbocharger coolant supply
2. Rear transfer tube

3. Heat shield vent lines
4. Heat shield
5. Turbocharger coolant return to bypass
6. Low temperature aftercooler vent line
7. Turbocharger vent line
8. Heat exchanger
9. Low temperature aftercooler coolant from heat exchanger
10. Sea water from heat exchanger
11. Priming line
12. Sea water inlet to heat exchanger
13. Sea water supply
14. Sea water pump
15. Low temperature aftercooler coolant to heat exchanger
16. Coolant from heat exchanger
17. Coolant bypass
18. Coolant pump
19. Aftercooler vent line
20. Aftercooler coolant supply
21. Aftercooler coolant return.

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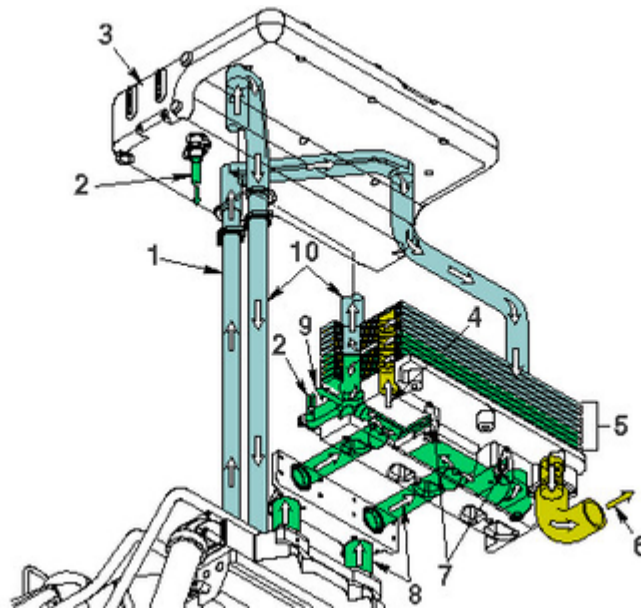
09600612

K38M Heat Exchanger Low Temperature Aftercooler- Plate Type

1. Turbocharger coolant supply
2. Rear transfer tube
3. Hat shield vent line
4. Heat shield
5. Turbocharger vent line
6. Turbocharger coolant return to bypass
7. Pressure cap
8. Heat exchanger plate pack
9. Sea water outlet

10. Priming line
11. Sea water inlet from sea water pump
12. Sea water supply
13. Sea water pump
14. Coolant bypass
15. Coolant from heat exchanger
16. Coolant pump
17. Aftercooler vent line
18. Aftercooler coolant supply
19. Aftercooler coolant return.

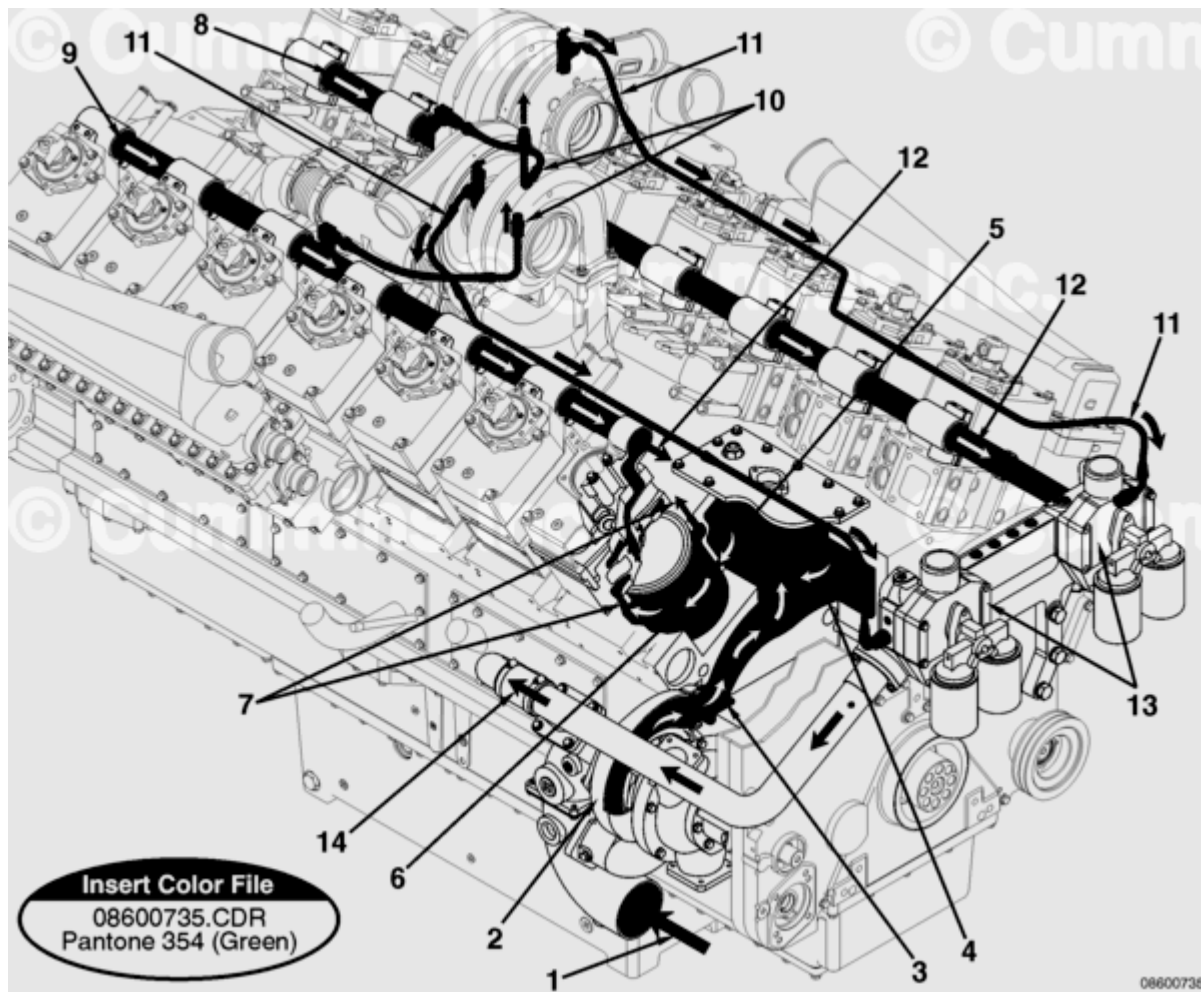
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03800612

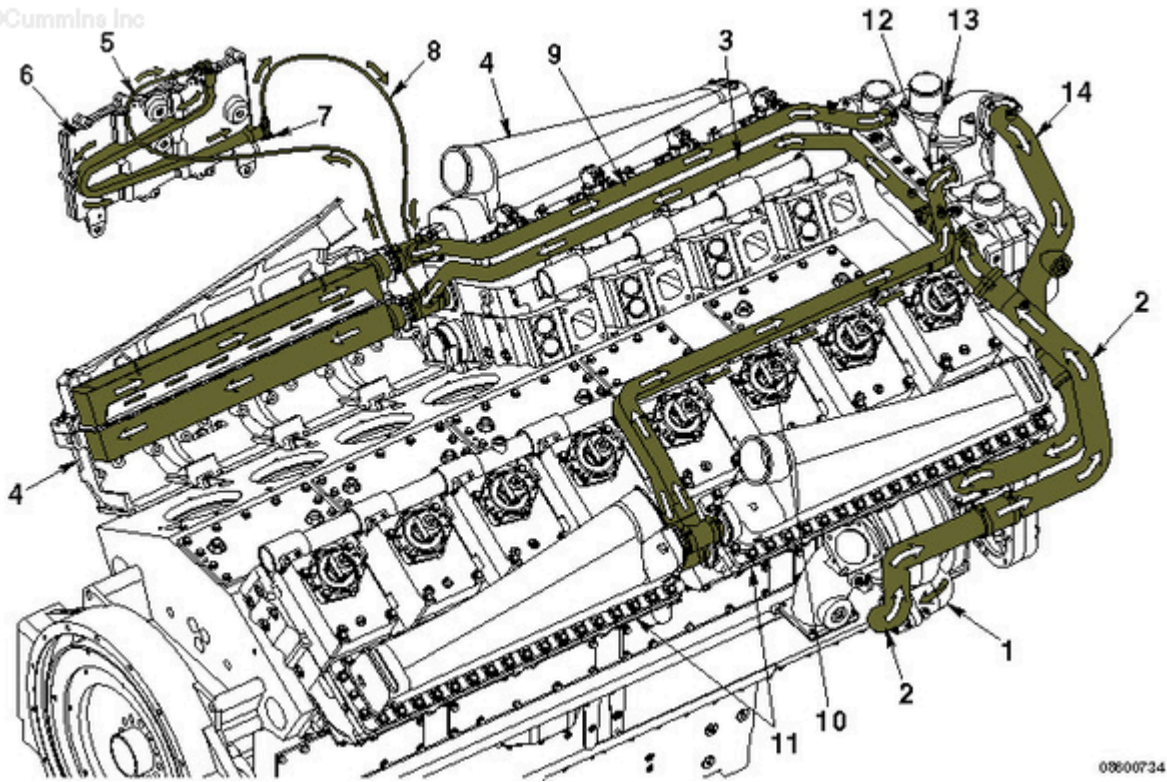
K38M and K50M Heat Exchanger - Plate Type

1. Low temperature aftercooler coolant from thermostat housing
2. Make-up tube
3. Expansion tank
4. Sea water inlet
5. Heat exchanger plate pack
6. Sea water outlet
7. Coolant filter ports
8. Engine coolant from thermostat housing
9. Engine coolant return to coolant pump
10. Low temperature aftercooler coolant return to thermostat housing.



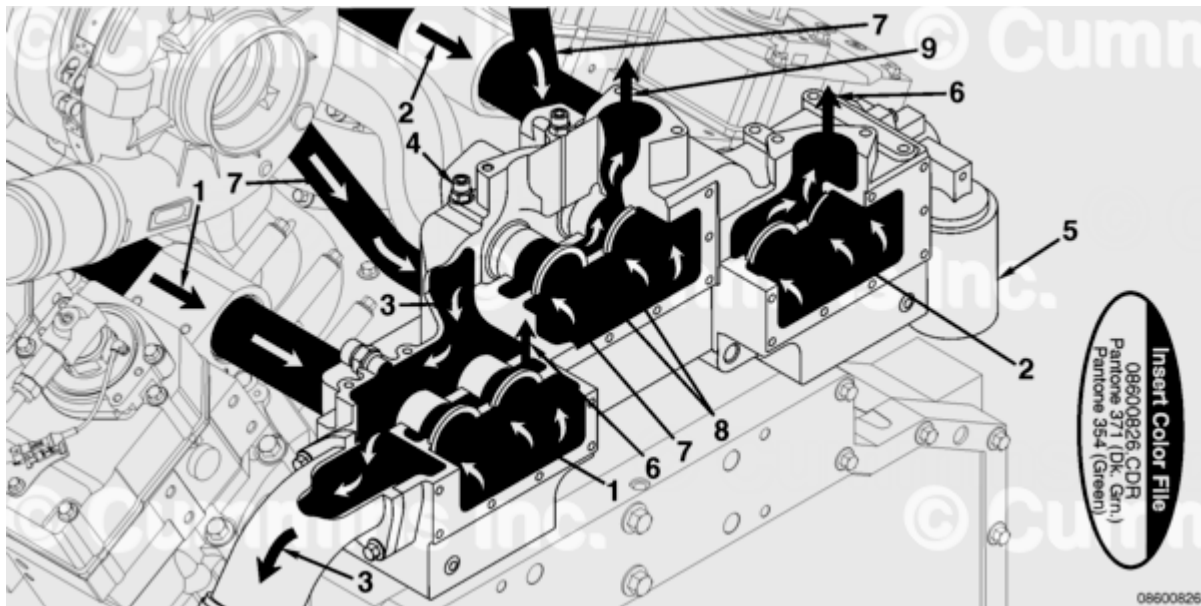
QSK50 Cooling System - with Electronically Actuated Injectors

1. Coolant inlet
2. Water pump
3. Coolant flow to oil cooler cavity
4. Coolant flow around oil coolers
5. Water jacket coolant
6. Coolant flow around cylinder liner
7. Coolant flow to cylinder head
8. Left bank water manifold
9. Right bank water manifold
10. Turbocharger coolant supply
11. Turbocharger coolant return to thermostat housing
12. Coolant flow to thermostat housing
13. Thermostat housing
14. Coolant bypass to water supply.



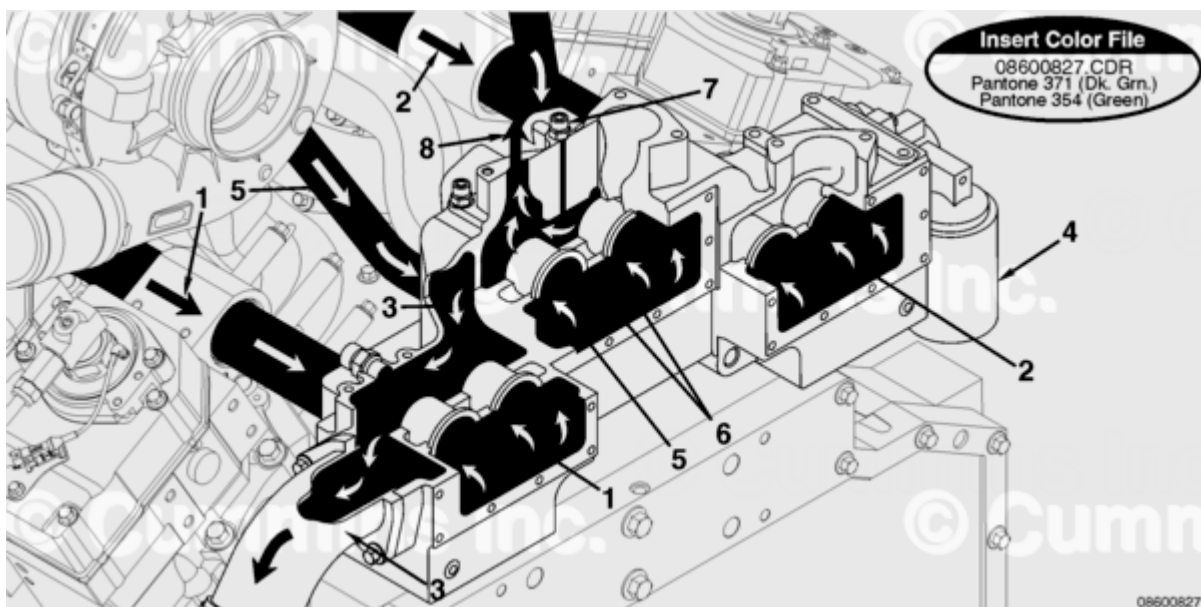
QSK50 Low Temperature Aftercooler System - with Electronically Actuated Injectors

1. Low temperature aftercooler coolant pump
2. Coolant supply
3. Left bank aftercooler coolant supply
4. Left bank aftercooler
5. Coolant supply to electronic control module (ECM) cooling plate
6. Electronic control module (ECM) cooling plate
7. Coolant drain valve
8. Electronic control module (ECM) coolant plate coolant return
9. Left bank aftercooler coolant return
10. Right bank aftercooler coolant supply
11. Right bank aftercooler
12. Right bank aftercooler coolant return
13. Low temperature aftercooler thermostat housing
14. Low temperature aftercooler coolant return.



QSK50 Industrial and Power Generation Thermostat Housing Coolant Flow (Thermostat Open)
- with Electronically Actuated Injectors

1. Right bank coolant to thermostat housing
2. Left bank coolant to thermostat housing
3. Coolant bypass to water jacket pump
4. Coolant vent port
5. Coolant filters
6. Coolant return to radiators
7. Low temperature aftercooler coolant from aftercooler
8. Low temperature aftercooler thermostats open
9. Low temperature aftercooler coolant supply to low temperature aftercooler radiator.



QSK50 Industrial and Power Generation Thermostat Housing Coolant Flow (Thermostat Closed) - with Electronically Actuated Injectors

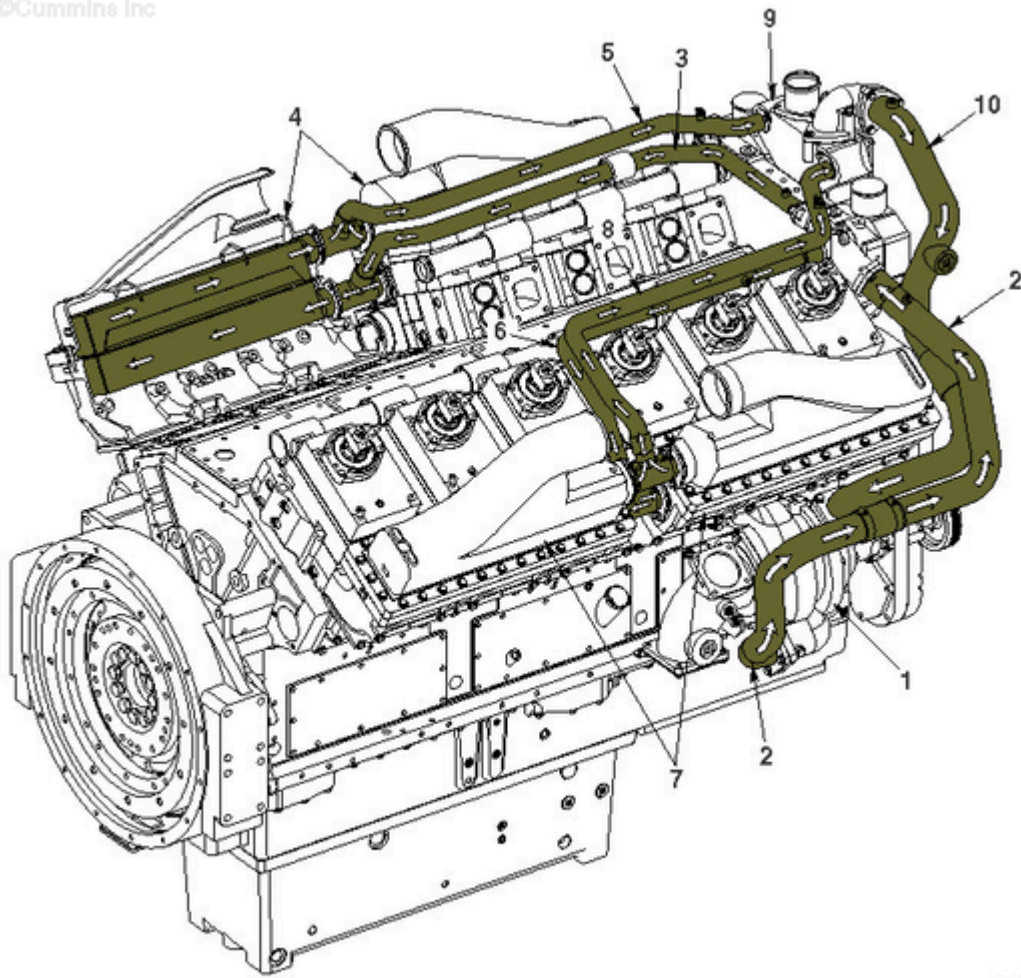
1. Right bank coolant to thermostat housing

2. Left bank coolant to thermostat housing
3. Coolant bypass to water jacket pump
4. Coolant filters
5. Low temperature aftercooler coolant from aftercooler
6. Coolant return to radiators
7. Low temperature aftercooler thermostats closed
8. Low temperature aftercooler vent.

13. Low temperature aftercooler make-up line to expansion tank

14. Low temperature aftercooler coolant return to low temperature aftercooler pump.

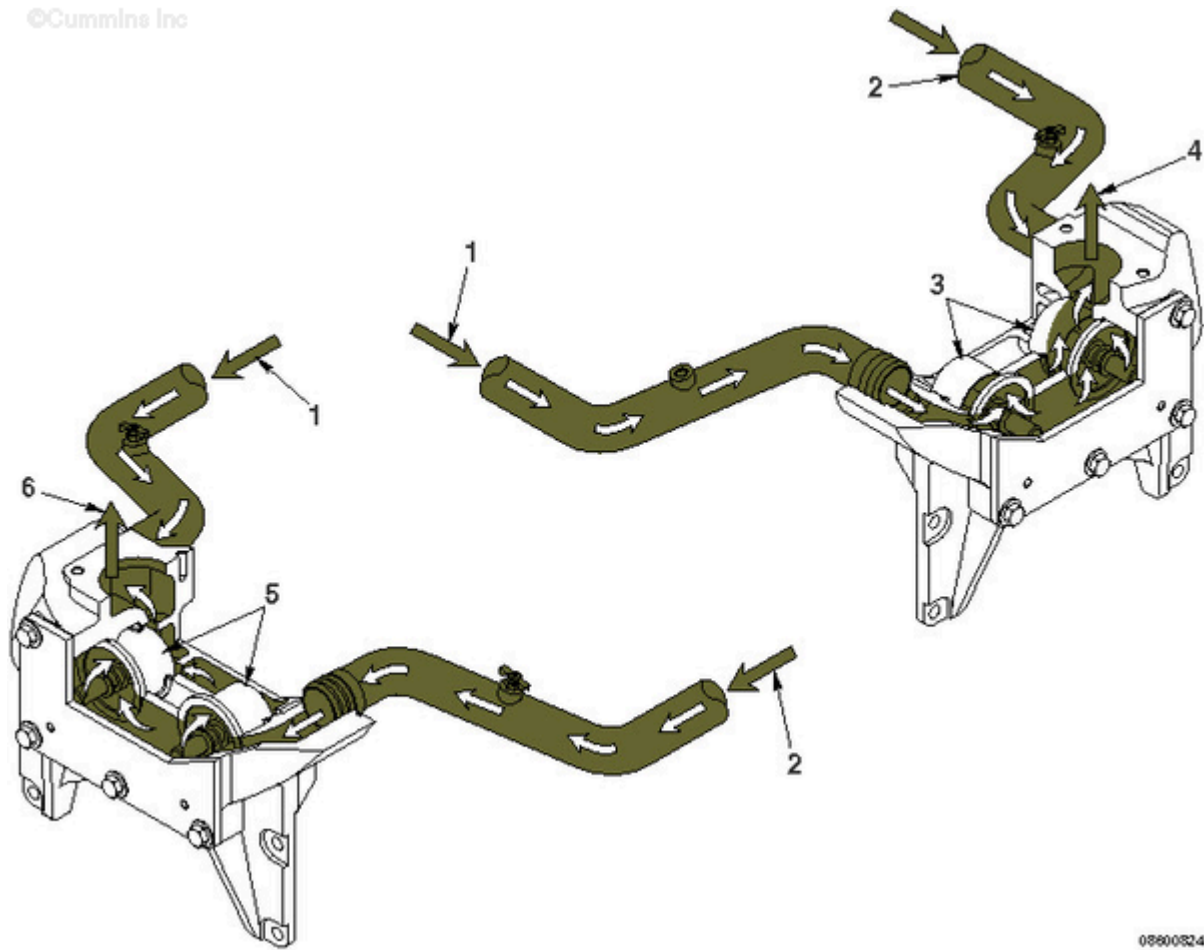
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08600812

QSK38 Industrial Low Temperature Aftercooler Coolant Flow

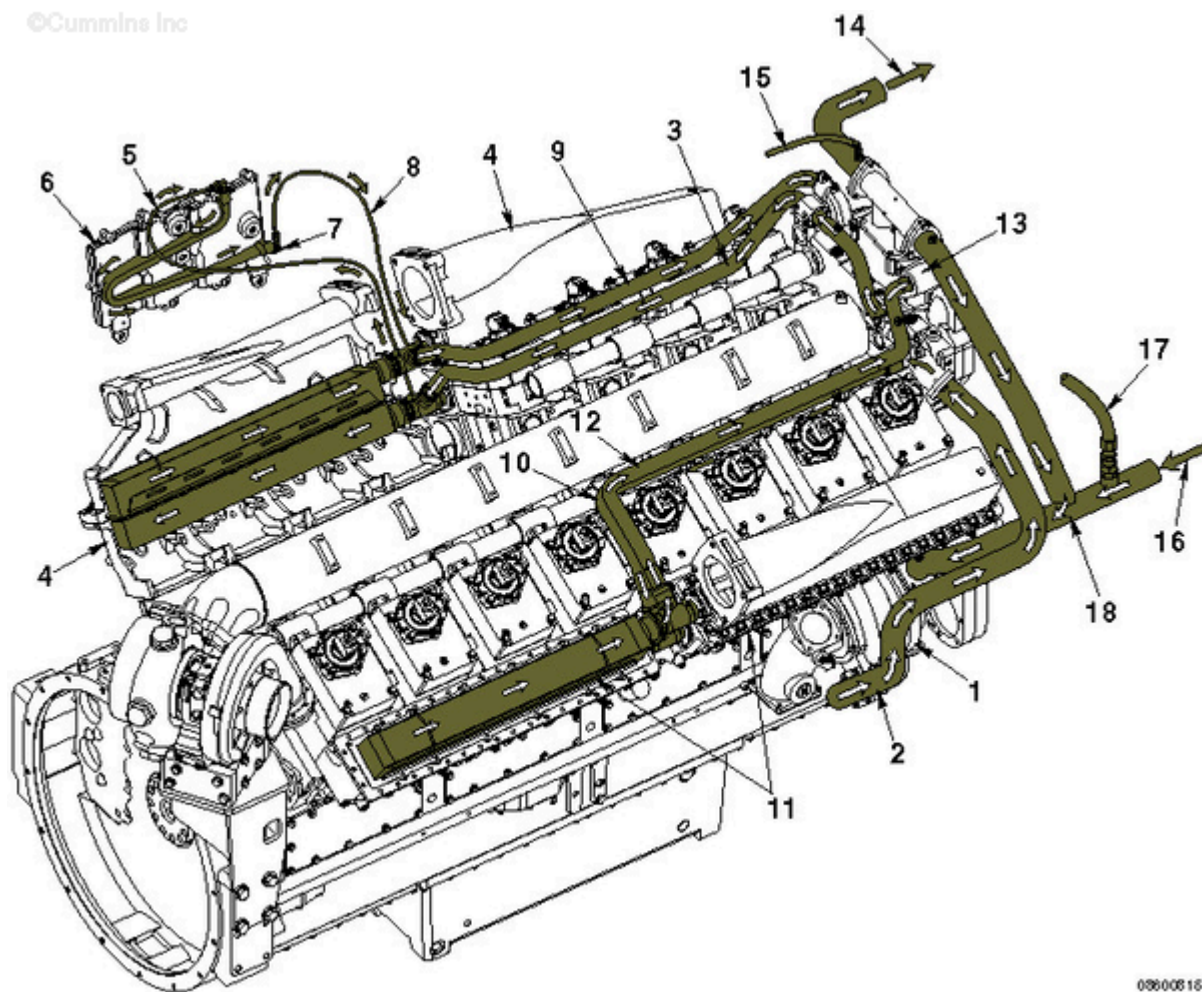
1. Low temperature aftercooler coolant pump
2. Low temperature aftercooler coolant supply
3. Left bank aftercooler coolant supply
4. Left bank aftercooler
5. Left bank aftercooler coolant return
6. Right bank aftercooler coolant supply
7. Right bank aftercooler
8. Right bank aftercooler return
9. Low temperature aftercooler thermostat housing
10. Low temperature aftercooler coolant return.



03800324

QSK38 and 50 Marine Low Temperature Aftercooler Coolant Flow

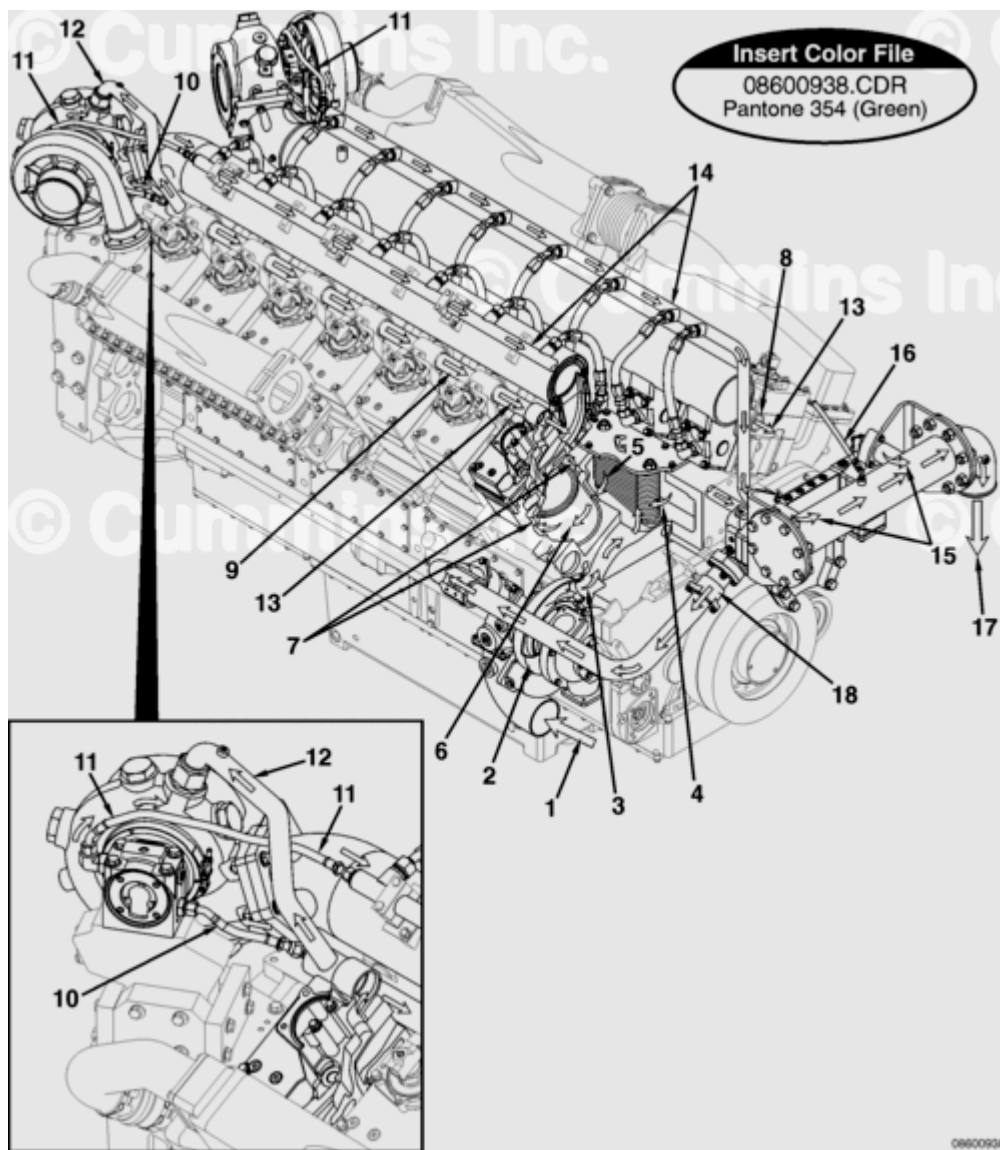
1. Right bank aftercooler coolant return to low temperature aftercooler thermostat housing
2. Left bank aftercooler coolant return to Low temperature aftercooler thermostat housing
3. Low temperature aftercooler thermostats open
4. Low temperature aftercooler coolant flow to keel cooler or heat exchanger
5. Low temperature aftercooler thermostats closed
6. Low temperature aftercooler coolant return to low temperature aftercooler pump.



09800815

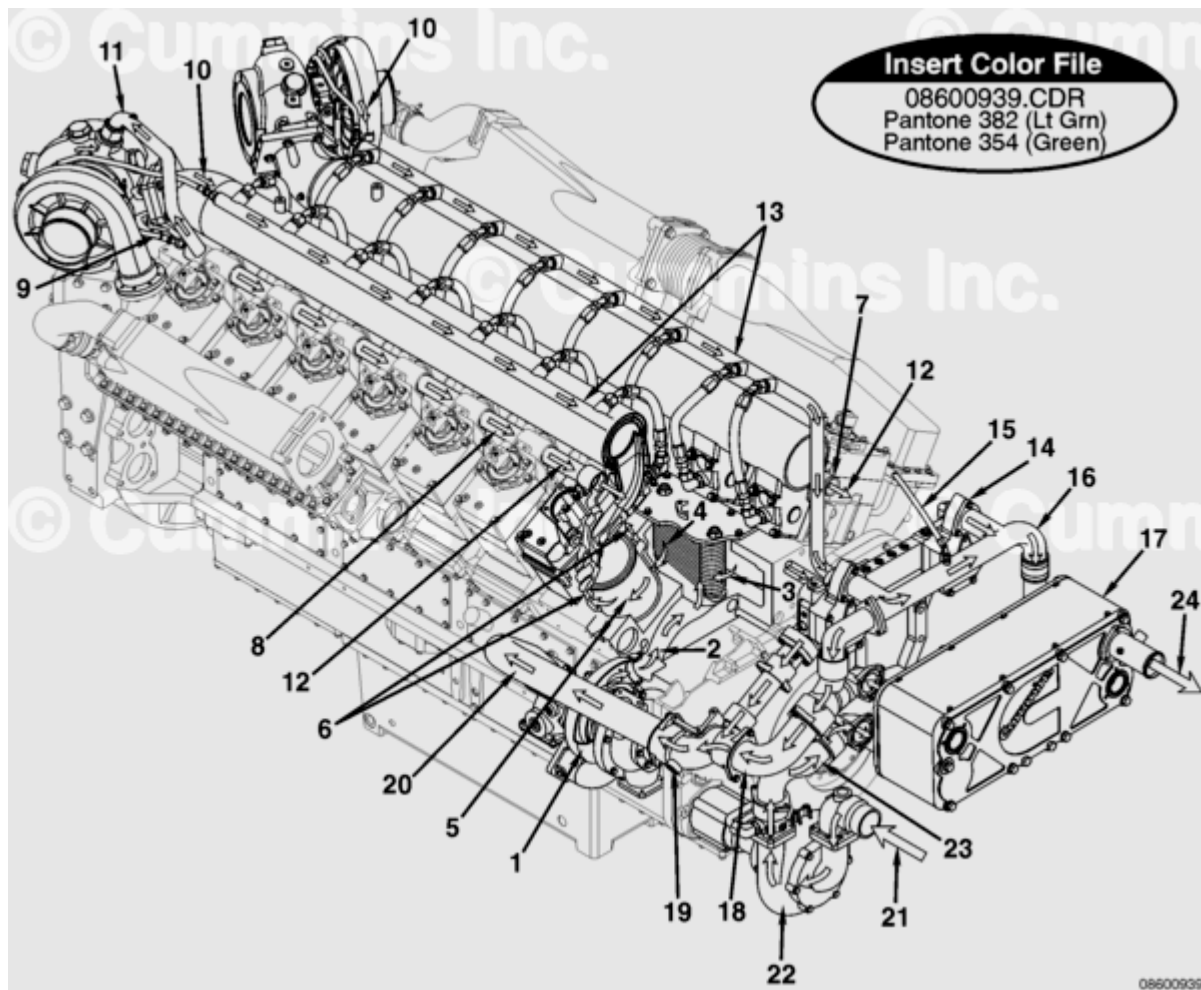
QSK50 Marine Low Temperature Aftercooler Coolant Flow

1. Low temperature aftercooler coolant pump
2. Low temperature aftercooler coolant supply
3. Left bank aftercooler coolant supply
4. Left bank aftercooler
5. Coolant supply to ECM coolant plate
6. ECM coolant plate
7. Coolant drain valve
8. ECM coolant plate coolant return
9. Left bank aftercooler coolant return
10. Right bank aftercooler coolant supply
11. Right bank aftercooler
12. Right bank aftercooler coolant return
13. Low temperature aftercooler thermostat housing
14. Low temperature aftercooler coolant flow to keel cooler or heat exchanger
15. Low temperature aftercooler vent line to expansion tank
16. Low temperature aftercooler coolant flow from keel cooler or heat exchanger
17. Low temperature aftercooler make-up line to expansion tank
18. Low temperature aftercooler coolant return to low temperature aftercooler pump.



QSK50 Marine Coolant (Keel Cooled)

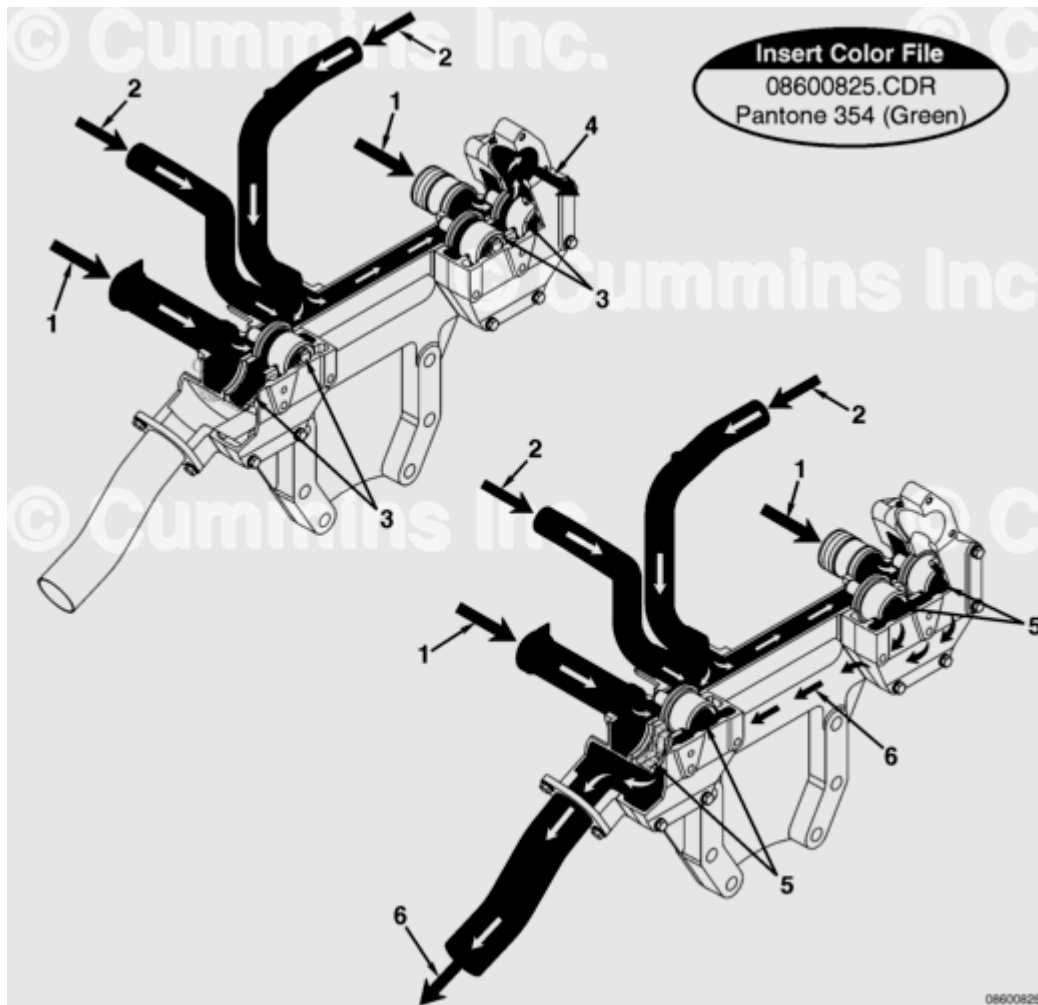
1. Coolant inlet
2. Water pump
3. Coolant flow to oil cooler cavity
4. Coolant flow around oil coolers
5. Water jacket coolant
6. Coolant flow around cylinder liner
7. Coolant flow to cylinder head
8. Left bank water manifold
9. Right bank water manifold
10. Turbocharger bearing housing coolant supply
11. Turbocharger bearing housing coolant return to exhaust manifold
12. Turbocharger coolant supply
13. Coolant flow from water manifold to thermostat housing
14. Coolant flow from exhaust manifold to thermostat housing
15. Coolant flow from thermostat housing
16. Vent line to expansion tank
17. Coolant flow to keel cooler
18. Coolant bypass to water pump.



QSK50 Marine Coolant (Heat Exchanger)

1. Water pump
2. Coolant flow to oil cooler cavity
3. Coolant flow around oil coolers
4. Water jacket coolant
5. Coolant flow around cylinder liner
6. Coolant flow to cylinder head
7. Left bank water manifold
8. Right bank water manifold
9. Turbocharger bearing housing coolant supply
10. Turbocharger bearing housing coolant return to exhaust manifold
11. Turbocharger coolant supply
12. Coolant flow from water manifold to thermostat housing
13. Coolant flow from exhaust manifold to thermostat housing
14. Thermostat housing
15. Vent line to expansion tank
16. Coolant flow to heat exchanger
17. Heat exchanger
18. Coolant flow from heat exchanger
19. Engine make-up line to expansion tank
20. Coolant bypass to water pump

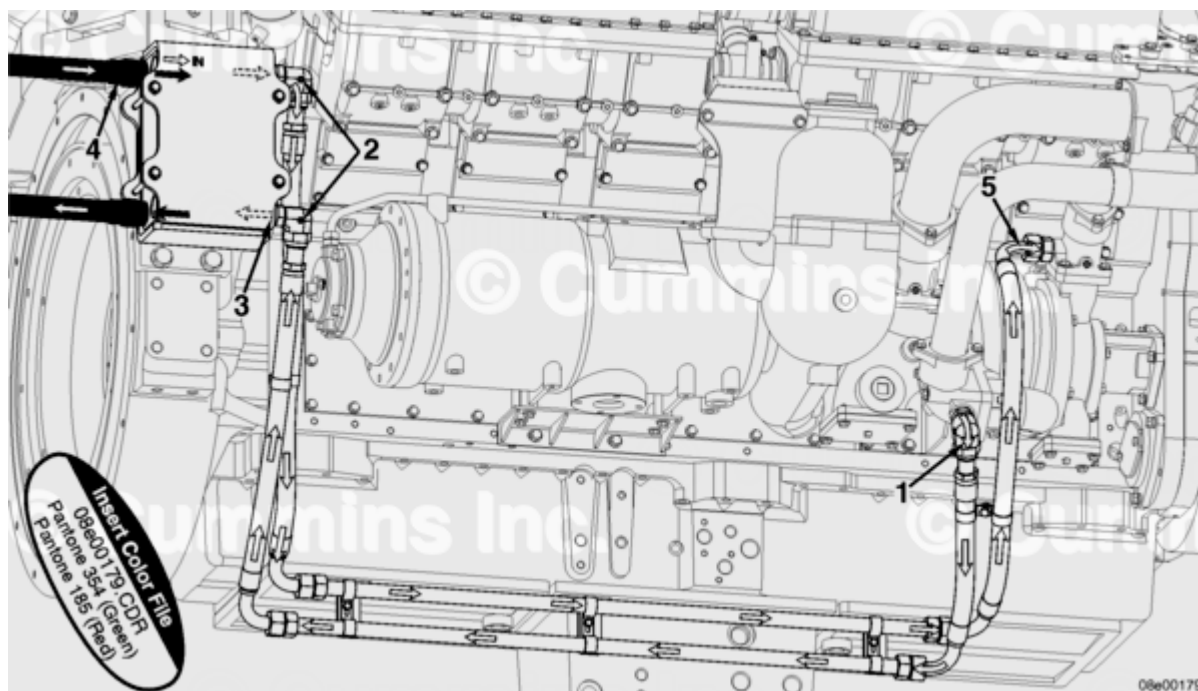
21. Sea water supply
22. Sea water pump
23. Sea water flow to heat exchanger
24. Sea water return.



Marine Thermostat Flow

1. Jacket water flow to thermostat housing
2. Exhaust manifold coolant flow to thermostat housing
3. Thermostats open
4. Coolant flow to keel cooler or heat exchanger
5. Thermostats closed
6. Coolant bypass to water jacket pump.

Note : QSK50 Marine engines with CM2150 and in a keel cooled configuration will have an insert in place of one thermostat. When facing the engine, the insert will be located in the left-most position in the thermostat housing. This insert will cause the flow to that position to always follow the closed thermostat path.



Marine Gear Oil Cooler

1. Coolant flow from LTA circuit
2. Marine gear oil cooler diagnostic ports (also used as air bleed ports for filling)
3. Marine gear oil cooler coolant inlet
4. Marine gear oil cooler oil inlet
5. Coolant return to LTA circuit.